Capstone Project Snapshot – JKLU

Project Title: Health Risk Analysis using ML & Environmental Data
Industry / Function: Environmental Health / Public Health/ Technology / AI & Machine Learning
Partner Organisation: JK Lakshmipat University (Academic Project)

Business Context:
• Indian cities like Delhi & Gurugram record hazardous AQI, causing serious respiratory and health risks
• No real-time forecasting or personalized health guidance for vulnerable groups

Project Objective:
• Real-time AQI forecasting with personalized health advisories for 29 Indian cities

Scope & Approach:
• CPCB dataset (2022–2025), 29 cities, 44 features
• CNN-BiLSTM + XGBoost; live APIs; Streamlit dashboard

Student Team & Mentorship:
• Priyanshi Mehta (2023Btech061), B.Tech CS & AI, JKLU
• Mentor: Dr. Amit Sinhal, CSE, IET, JKLU

Project Duration:
• Minor Project, AY 2025–26 (~16 Weeks)

Key Deliverables:
• CNN-BiLSTM & XGBoost models for 29 cities; Streamlit dashboard; project report.

Business Impact / Key Insights:
• CNN-BiLSTM: $R^2 > 0.98$; outperforms XGBoost for longer horizons
• Gurugram most polluted (avg. AQI 400); PM2.5 & PM10 dominant
• Crop burning spikes AQI; personalized advisories bridge forecasting with health action

Potential Replication for JKO Companies:
• Smart city air quality monitoring platforms for municipal corporations
• Employee health & safety advisory tools for industries in high-pollution zones

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